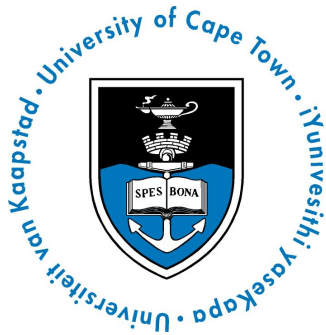


University of Cape Town



Supervised Learning Assignment 2

AirBnB Bookings Prediction

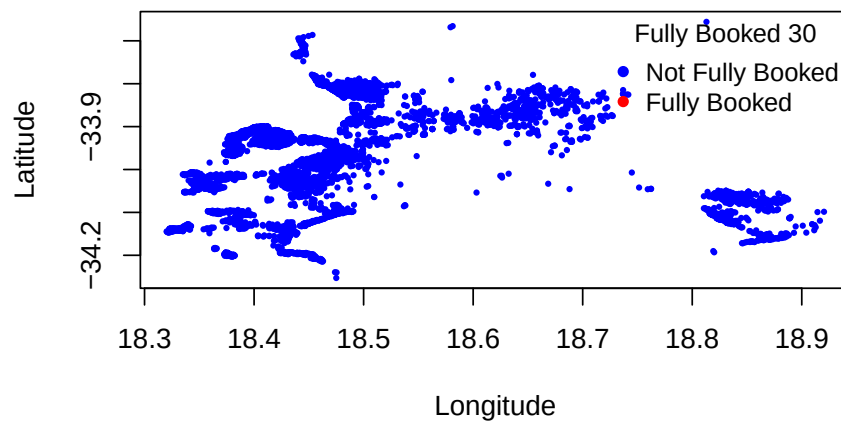
Pamela Afful

Student Number: AFFPAM001

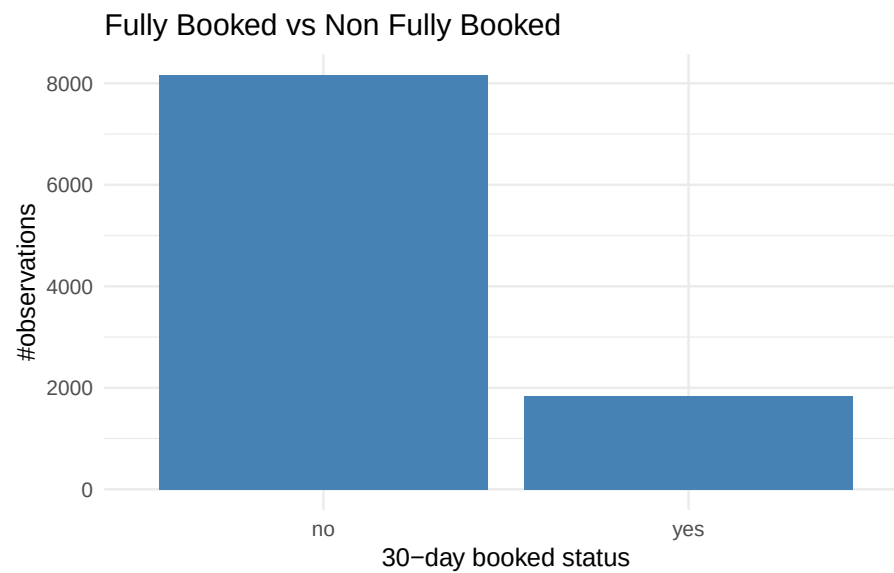
May 7, 2026

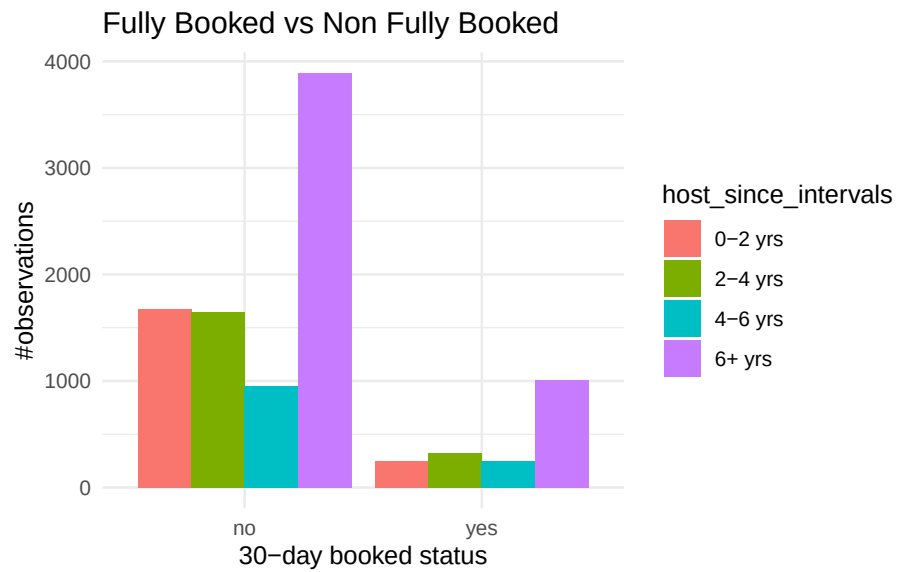
You can add options to executable code like this

Data Wrangling



EDA and Feature Selection





Use a novel decision tree to gauge which features could be important for determining the response variable. Since we have an imbalanced class, a weighted decision tree is used:

<https://machinelearningmastery.com/cost-sensitive-decision-trees-for-imbalanced-classification/>

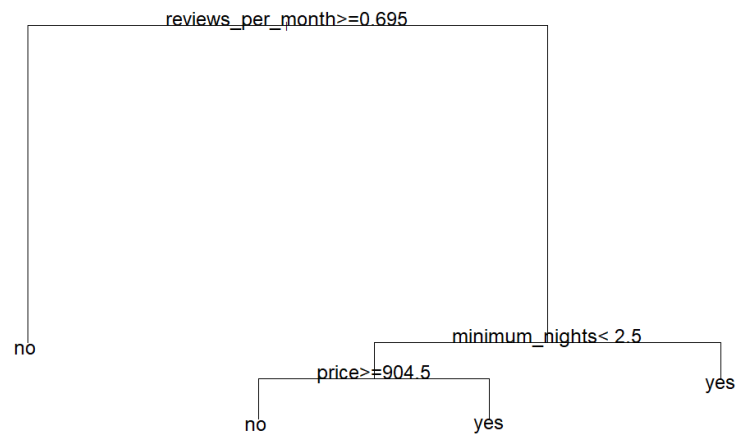
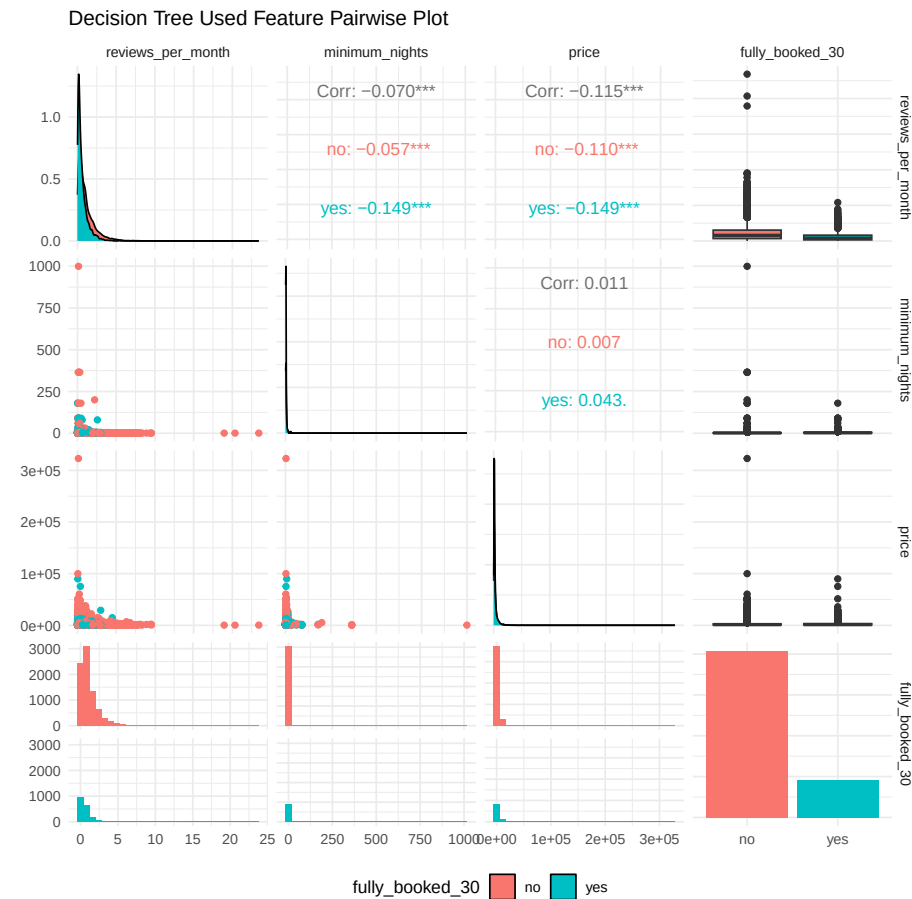


Figure 1: Initial Decision Tree Feature/ Class Splits

Features used to reduce RSS during splitting were **reviews_per_month** (the largest reduction), **minimum_night**, and **price**.

BnB listings with reviews_per_month greater than 0.695 all had terminal nodes with the categorisation “no” for the response variable. Price and minimum nights were determining factors for listings with a minimum number of nights less than 2.5, which ended in “yes”. Further splitting this region, for listings where the minimum was greater than 2.5, listings with prices greater than 905.5 were also categorised as “yes”.



- comeback and add more

Model Training

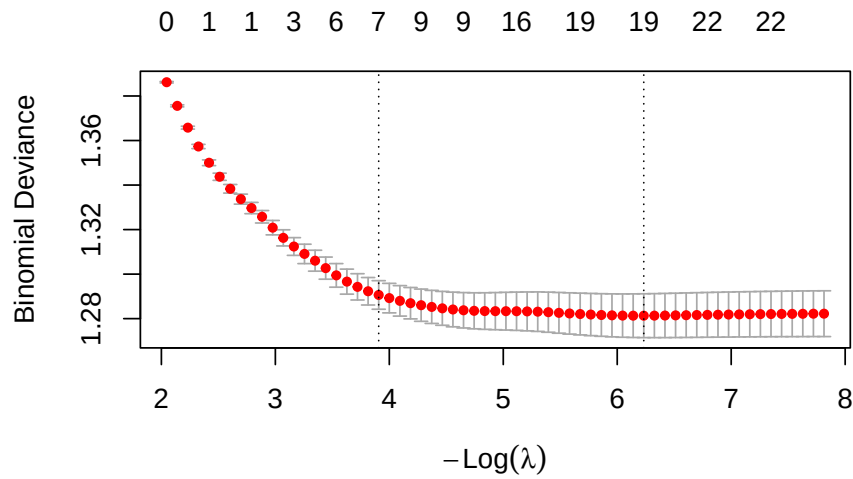
Baseline Model Vanilla Logistic Regression

Characteristic	log(OR) ^f	SE
(Intercept)	26*	10.2
host_response_time		
within an hour	—	—
within a few hours	0.18*	0.073
within a day	0.42***	0.097
a few days or more	0.47	0.324
host_response_rate	0.00	0.003
host_acceptance_rate	0.00*	0.002
host_is_superhost		
t	—	—
f	0.07	0.060
host_listings_count	-0.01***	0.001
host_verifications		
phone_only	—	—
phone_email	0.05	0.097
email_only	2.0	1.50
latitude	0.56*	0.278
longitude	-0.54*	0.252
room_type		
Entire home/apt	—	—
Private room	-0.15	0.103
accommodates	-0.01	0.030
bathrooms	-0.08	0.047
bedrooms	0.12*	0.056
beds	0.00	0.027
price	0.00	0.000
minimum_nights	-0.01	0.011
maximum_nights	0.00	0.000
minimum_nights_avg_ntm	0.02	0.011
maximum_nights_avg_ntm	0.00	0.000
number_of_reviews	0.00	0.001
reviews_per_month	-0.67***	0.054
review_scores_rating	0.23**	0.085
instant_bookable		
t	—	—
f	0.21**	0.073
host_since_months_int	0.00***	0.001
^f *p<0.05; **p<0.01; ***p<0.001		
Abbreviations: CI = Confidence Interval, OR = Odds Ratio, SE = Standard Error		
No. Obs. = 10,000		

Figure 2: Baseline Logistic Regression Summary

-
- Significant Features:
 - Intercept (which absorbes the omitted categorical variables: within a few hours, instant bookable=T)
 - Latitude and Longitude,

- Bedrooms,
- Reviews per month
- Review Score Ratings
- Instant bookable=F,
- Host Since (Measured in months)
- Insignificant Features:
 - The rest* comeback
 - Most interestingly the price!-

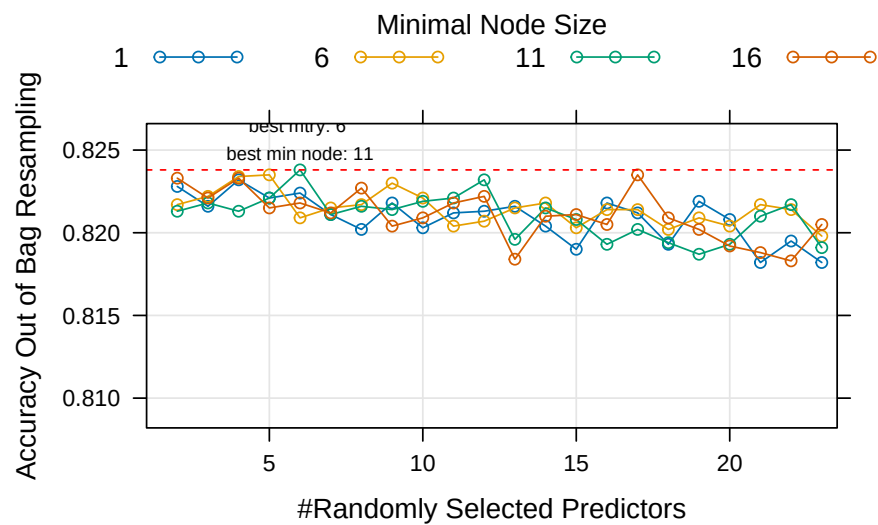


From the above, the best number of non-zero coefficients equals 9 (using 1SE discrimination). This is used to identify the zero-coefficient features.

Penalized Features:

- response ratwe
- acceptance rate
- is_superhost
- host ver
- lat and long
- all those in red

Feature	Coefficient
(Intercept)	-0.0717522
host_response_time	0.0997685
host_response_rate	0.0000000
host_acceptance_rate	0.0000000
host_is_superhost	0.0000000
host_listings_count	-0.0042548
host_verifications	0.0000000
latitude	0.0000000
longitude	0.0000000
room_type	0.0000000
accommodates	0.0000000
bathrooms	0.0000000
bedrooms	0.0000000
beds	0.0000000
price	0.0000000
minimum_nights	0.0000000
maximum_nights	0.0000000
minimum_nights_avg_ntm	0.0073820
maximum_nights_avg_ntm	0.0000000
number_of_reviews	0.0000000
reviews_per_month	-0.5037625
review_scores_rating	0.0183948
instant_bookable	0.0868937
host_since_months_int	0.0013796



```

      mtry splitrule min.node.size
49      6      gini           11
[1] 0.8238

```

Boosting Model

Used non-zero features from the regularised model to speed up training- we've established that the majority of these are not relevant anyway through CV

Used features:

-